

Joint symmetry and dynamical accessibility in compact Hamiltonian encodings of set cover

Fabrício de Souza Luiz

Instituto de Física Gleb Wataghin, Universidade Estadual de Campinas, 13083-859 Campinas, SP, Brazil
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Which part of a Hamiltonian spectrum is physically relevant when both the initial state and the interpolation preserve several symmetries? We answer this question for a compact multi-register encoding of Minimum Set Cover. The represented symmetry is the direct product of register permutations with the faithful base action of the incidence automorphism group. We distinguish its fixed, symmetry-allowed space from the generally smaller cyclic space generated by the protocol, and define gaps relative to an isolated band in that cyclic space. A sector-resolved Schur-complement bound certifies solution probability in sectors that contain feasible covers and have positive invalid-state separation; unlike the global statement, it necessarily contains the sector's kinetic floor. Exact orbit quotients expose crossings that close a global multiplicity gap while remaining dark to the symmetric protocol; a stability theorem shows that transverse dark crossings persist under small symmetry-preserving perturbations. For an even-cycle family, the original linear interpolation has an exact global crossing with a constant gap in the joint fixed sector. We also give an explicit Johnson/Metropolis parent-Hamiltonian path with a Dicke ground state at one endpoint and cover probability $1 - O(n^{-5})$ at the other. A zero-range comparison reduces its accessible-gap problem to a three-box heat-bath chain. A log-concave coupling proves the uniform bound $\Delta_{\text{acc},\min} = \Omega(n^{-13})$; exact quotients display much larger finite-size gaps but are not used as an asymptotic proof. In the abstract Hamiltonian-evolution model, derivative bounds also give a polynomial adiabatic runtime, conditional on Dicke-state preparation and Hamiltonian access. The resulting obstruction is conditional, not an initial-state-independent no-go: a fixed closed Hamiltonian cannot relax, and preserved symmetries forbid transitions between sectors, but a different state preparation or a symmetry-breaking generator changes the accessible spectrum. We

make no quantum-speedup claim; the contribution is a rigorous separation of global, symmetry-allowed, and dynamically accessible spectral structure.

1 Introduction

Symmetry block diagonalizes a Hamiltonian, but block diagonalization alone does not determine which eigenstates a protocol can occupy. If an initial state is fixed by every preserved symmetry, evolution is confined to the joint fixed space. Even that statement can be loose: the algebra generated by the endpoint Hamiltonians may act cyclically only on a proper subspace of the joint fixed space. A small global gap can then describe a crossing that the protocol neither couples to nor detects.

Relation to prior work. Set Cover admits the usual one-binary-variable-per-base Ising formulation, and it also appears among the constrained examples motivating quantum alternating-operator ansatzes [1, 2]. Separately, qubit-efficient variational encodings show that logarithmic or otherwise compressed register counts are not by themselves a novelty claim [3–5]. Our multi-register address encoding belongs to this broad compact-encoding line; the contribution claimed here is spectral rather than a new qubit count.

Classical objective symmetries can confine QAOA dynamics and impose equal measurement probabilities on symmetry-related strings [6]. Quantum walks on automorphism quotients likewise identify a smaller invariant evolution for suitable initial states [7], while Krylov methods use repeated generator action to identify a minimal dynamical subspace [8]. These ideas motivate, but do not identify with one another, the two spaces used below: the group-fixed orbit space and the generally smaller cyclic space of the full protocol algebra.

Finally, symmetric discriminants of reversible Markov chains and annealing paths with Gibbs-amplitude ground states are established tools in quantum walk and quantum simulated-annealing constructions [9, 10]. Quantitative adiabatic statements require an isolated spectral band together with derivative and gap control [11]. We use the discriminant

only to construct a rigorously gapped witness path and make no speedup claim.

Minimum Set Cover (MSC) provides a concrete setting in which these three spaces differ. A k -register encoding represents an ordered candidate by $k\lceil\log_2 n_B\rceil$ address qubits. Permuting the registers gives an S_k action, while colour-preserving automorphisms of the incidence relation permute the base labels. Register symmetry alone and a final-Hamiltonian sector gap are insufficient for an adiabatic or otherwise path-dependent statement: one must use the faithful joint action, the protocol's cyclic space, and the minimum isolation gap along an explicit path.

This rewrite makes that distinction its central question. Its main results are:

1. a representation-theoretic definition of the effective group $S_k \times G_B$, together with exact orbit quotients and a strict separation between the symmetry-allowed space $\mathcal{H}_{\text{sym}}$ and the cyclic space $\mathcal{K}_u$;
2. a sector-resolved localization theorem in which the separation from invalid states contains the kinetic floor of the chosen sector;
3. a protocol-dependent taxonomy separating ground multiplicity, the gap above a manifold, the isolation gap of a tracked band, and the energy penalty caused by inaccessibility;
4. finite and asymptotic examples of dynamically dark crossings, with a stability theorem for transverse symmetry-preserving perturbations; and
5. an explicit parent path on an even-cycle family, with exact endpoint concentration and a uniform polynomial accessible-gap certificate obtained from a log-concave three-box coupling, together with a conditional polynomial-time adiabatic corollary.

The last result is purposely about accessibility rather than computational advantage. The controlled cycle family is classically easy, and the proven exponent is conservative. Its role is to separate the gap question cleanly into symmetry, geometry, and kinetics, without inferring asymptotic behavior from finite-size data.

Accordingly, neither orbit reduction, compact addressing, nor a Markov discriminant is claimed as new in isolation. The new combination is the three-way separation of global, group-fixed, and protocol-cyclic spectra for a multi-register coverage Hamiltonian; the kinetic-floor localization bound inside a symmetry sector; and an analytic family in which an exact global inter-irrep crossing is dynamically dark while a distinct parent protocol has a uniformly polynomial cyclic-gap certificate.

2 Encoding, joint symmetry, and cyclic accessibility

Let $A \in \{0,1\}^{n_B \times n_T}$ be the incidence matrix, with $A_{bt} = 1$ when base b covers task t . We first work in the physical address space

$$\mathcal{H}_{\text{valid}} = (\mathbb{C}^{n_B})^{\otimes k}; \quad (1)$$

the embedding into $2^{\lceil\log_2 n_B\rceil}$ levels and its dynamically disconnected padding block are implementation details.

For a basis tuple $\mathbf{b} = (b_1, \dots, b_k)$, let $U(\mathbf{b})$ be the number of uncovered tasks and $D(\mathbf{b})$ the number of equal register pairs. The problem Hamiltonian is

$$H_{\text{prob}} = H_{\text{walk}} + \lambda H_{\text{cov}} + \mu H_{\text{excl}}, \quad (2)$$

where the last two terms are diagonal with eigenvalues U and D . The single-register complete-graph walk is normalized by $1/(n_B - 1)$. An explicit initial Hamiltonian used for the linear interpolation is

$$H_{\text{init}} = \sum_{r=1}^k (I - |u_B\rangle\langle u_B|)_r, \quad |u_B\rangle = n_B^{-1/2} \sum_{b=1}^{n_B} |b\rangle, \quad (3)$$

with unique ground state $|u\rangle = |u_B\rangle^{\otimes k}$ and initial gap one.

2.1 The faithfully represented group

The colour-preserving incidence automorphism group is

$$G_A = \{(\pi_B, \pi_T) : A_{\pi_B(b), \pi_T(t)} = A_{bt}\}. \quad (4)$$

Only its base component acts on Eq. (1). We therefore use the effective group

$$G_B = \text{pr}_B(G_A) \simeq G_A / \ker \rho, \quad \rho(\pi_B, \pi_T) = \pi_B, \quad (5)$$

rather than treating a possibly non-faithful G_A representation as if all its irreps could occur. The register and base actions commute, so the represented symmetry is

$$G = S_k \times G_B. \quad (6)$$

Every term in Eqs. (2) and (3) commutes with G . Consequently

$$\mathcal{H}_{\text{valid}} \simeq \bigoplus_{\alpha, \beta} V_{\alpha}^{S_k} \otimes V_{\beta}^{G_B} \otimes \mathcal{M}_{\alpha\beta}, \quad (7)$$

$$H \simeq \bigoplus_{\alpha, \beta} I_{V_{\alpha}} \otimes I_{V_{\beta}} \otimes h_{\alpha\beta}. \quad (8)$$

The second identity is the source of Schur multiplicities; an equality of eigenvalues belonging to different blocks is a separate phenomenon.

2.2 Allowed space versus accessible space

The joint fixed projector and its range are

$$\Pi_{\text{sym}} = \left(\frac{1}{k!} \sum_{\sigma \in S_k} U_{\sigma} \right) \left(\frac{1}{|G_B|} \sum_{g \in G_B} V_g \right), \quad (9)$$

$$\mathcal{H}_{\text{sym}} = \text{Ran } \Pi_{\text{sym}}. \quad (10)$$

This is the symmetry-allowed invariant space, not automatically the space filled by the protocol. Let $\mathfrak{A} = \text{alg}^*(H_{\text{init}}, H_{\text{prob}}, I)$ be the unital star-algebra generated by the endpoints. The dynamically accessible cyclic space is

$$\mathcal{K}_u = \mathfrak{A}|u\rangle \quad (11)$$

$$= \text{span}\{A_m \cdots A_1|u\rangle : A_i \in \{H_{\text{init}}, H_{\text{prob}}\}\}. \quad (12)$$

It is the smallest common reducing subspace that contains the initial state, and in general

$$\mathcal{K}_u \subseteq \mathcal{H}_{\text{sym}} \subseteq \mathcal{H}_{\text{valid}}. \quad (13)$$

The first inclusion can be strict; padding makes it strict automatically if one defines the symmetry on the full qubit space.

2.3 Exact orbit quotient

Orbit quotients of symmetry-confined quantum walks are established constructions [7]; we use the joint action here to retain the exact Hamiltonian matrix elements relevant to the present encoding. For a G -orbit $\mathcal{O}$ in the computational basis, define $|\mathcal{O}\rangle = |\mathcal{O}|^{-1/2} \sum_{x \in \mathcal{O}} |x\rangle$. These orbit states are an orthonormal basis of $\mathcal{H}_{\text{sym}}$. If H is invariant and $x \in \mathcal{O}$, then

$$\langle \mathcal{O}' | H | \mathcal{O} \rangle = \sqrt{\frac{|\mathcal{O}|}{|\mathcal{O}'|}} \sum_{y \in \mathcal{O}'} H_{yx}. \quad (14)$$

The sum is independent of the representative. Burnside's lemma gives

$$\dim \mathcal{H}_{\text{sym}} = \frac{1}{|G_B|} \sum_{g \in G_B} [z^k] \prod_{\ell \geq 1} (1 - z^{\ell})^{-c_{\ell}(g)}, \quad (15)$$

where $c_{\ell}(g)$ counts length- ℓ cycles in the base action. Orbit states need not span $\mathcal{K}_u$ minimally; cyclic closure must be tested separately.

3 Sector-resolved localization

Let P project onto the span of computational-basis tuples that use distinct bases and cover every task, and put $Q = I - P$. Write $V = \lambda H_{\text{cov}} + \mu H_{\text{excl}}$, $m = \min\{\lambda, \mu\}$, and $\kappa = k/(n_B - 1)$. Then $VP = 0$ and the complete-graph normalization gives

$$QH_{\text{prob}}Q \geq (m - \kappa)Q. \quad (16)$$

The global localization statement cannot be copied verbatim into an arbitrary sector. A projected cover is a superposition of computational covers and can have nonzero kinetic expectation. The missing term is the sector's kinetic floor.

Proposition 1 (Sector localization). *Let Π_{γ} be a symmetry-sector projector commuting with H_{prob} , H_{walk} , V , and P , and suppose $P_{\gamma} = P\Pi_{\gamma} \neq 0$. Define*

$$Q_{\gamma} = Q\Pi_{\gamma}, \quad B_{\gamma} = Q_{\gamma}H_{\text{walk}}P_{\gamma}, \quad (17)$$

$$\eta_{\gamma} = \lambda_{\min}(P_{\gamma}H_{\text{walk}}P_{\gamma}|_{\text{Ran } P_{\gamma}}). \quad (18)$$

Let E_0^{γ} be the lowest energy in the sector and

$$\Gamma_{\gamma} = \lambda_{\min}(Q_{\gamma}H_{\text{prob}}Q_{\gamma}) - E_0^{\gamma}. \quad (19)$$

If $\Gamma_{\gamma} > 0$, every normalized sector ground state ψ_{γ} satisfies

$$1 - \langle \psi_{\gamma} | P | \psi_{\gamma} \rangle \leq \frac{\|B_{\gamma}\|^2}{\Gamma_{\gamma}^2 + \|B_{\gamma}\|^2}. \quad (20)$$

Moreover,

$$\Gamma_{\gamma} \geq m - \kappa - \eta_{\gamma}, \quad (21)$$

so the right-hand side of Eq. (20) remains valid with Γ_{γ} replaced by $g_{\gamma} = m - \kappa - \eta_{\gamma} > 0$.

Proof. The variational principle applied inside $\text{Ran } \Pi_{\gamma}$ gives $E_0^{\gamma} \leq \eta_{\gamma}$. Combining this with Eq. (16) proves Eq. (21). For a sector ground vector, write $p = P_{\gamma}\psi_{\gamma}$ and $q = Q_{\gamma}\psi_{\gamma}$. The projected eigenvalue equation is

$$(Q_{\gamma}H_{\text{prob}}Q_{\gamma} - E_0^{\gamma})q = -B_{\gamma}p. \quad (22)$$

The inverse on the left has norm at most Γ_{γ}^{-1} , hence $\|q\| \leq \|B_{\gamma}\|\|p\|/\Gamma_{\gamma}$. Using $\|p\|^2 + \|q\|^2 = 1$ gives Eq. (20). $\square$

This statement certifies the solution content of the lowest-energy state within a sector. It does not say that the ground vector has exact support on $\text{Ran } P$: for finite penalties Eq. (20) controls, rather than eliminates, leakage. Exact support occurs only in an infinite penalty limit or in special decoupled cases.

There is also a cyclic-space caveat. The projector onto $\mathcal{K}_u$ need not commute with P , so $P\Pi_{\mathcal{K}}$ need not be an orthogonal projector. Proposition 1 must be used for genuine symmetry sectors; cyclic-space solution probability is then evaluated directly or bounded through a separately constructed commuting projector.

4 Protocol-dependent gaps and conditional no-go statements

For an invariant space $\mathcal{L}$, let $P_0^{\mathcal{L}}(s)$ be the ground projector of $H(s)|_{\mathcal{L}}$ and $g_0^{\mathcal{L}}(s) = \text{rank } P_0^{\mathcal{L}}(s)$. Ordering

eigenvalues with multiplicity, we keep separate

$$\delta_{\text{mult}}^{\mathcal{L}}(s) = E_1^{\mathcal{L}}(s) - E_0^{\mathcal{L}}(s), \quad (23)$$

$$\Delta_{\text{exc}}^{\mathcal{L}}(s) = E_{g_0^{\mathcal{L}}(s)}^{\mathcal{L}}(s) - E_0^{\mathcal{L}}(s). \quad (24)$$

The first vanishes whenever the ground state is degenerate; the second is the excitation gap above the whole manifold. A Schur multiplicity or an exact coincidence between irreps is therefore not called a gap closure when Eq. (24)'s second quantity stays positive.

The nested spaces in Eq. (13) define energies E_0 , E_0^{sym} , and $E_0^{\mathcal{K}}$ and the accessibility costs

$$\delta_{\text{sym}} = E_0^{\text{sym}} - E_0, \quad \delta_{\text{cyc}} = E_0^{\mathcal{K}} - E_0^{\text{sym}}, \quad (25)$$

$$\delta_{\text{acc}} = E_0^{\mathcal{K}} - E_0 = \delta_{\text{sym}} + \delta_{\text{cyc}}. \quad (26)$$

A positive δ_{acc} says that the protocol follows a globally excited branch. That branch can nevertheless have high cover probability.

For adiabatic evolution, the relevant object is an isolated spectral band, not merely the first distinct final eigenvalue [11]. Let $P_{\text{track}}(s) \subset \mathcal{K}_u$ be a continuous isolated spectral projector containing the initial ground state. Define

$$\Delta_{\text{ad}}(s) = \text{dist}(\text{spec } H_{\text{track}}(s), \text{spec } H_{\mathcal{K}_u}(s) \setminus \text{spec } H_{\text{track}}(s)). \quad (27)$$

An isolated continuous projector has constant rank. If its intended rank changes, an isolation gap has closed and a nondegenerate adiabatic formula cannot simply be continued through that point.

What fixed-Hamiltonian evolution does not prove. For a time-independent Hamiltonian $H = \sum_j E_j P_j$, unitary evolution preserves every energy-eigenspace population,

$$\|P_j \psi(t)\|^2 = \|P_j \psi(0)\|^2. \quad (28)$$

This rules out closed-system relaxation to a different eigenstate and fixes the diagonal ensemble. It does *not* imply that the probability of a computationally marked subspace is constant: interference between distinct energies can create transient concentration. A finite time scan cannot prove that no later time does better.

The statement is also initial-state dependent. A solution eigenstate used as the initial state already has unit solution population; a state with appropriate coherences can reach a marked subspace transiently. The exact symmetry obstruction is narrower and stronger:

Lemma 2 (Symmetry-conditioned inaccessibility). *If $[H(t), \Pi_\gamma] = 0$ for every t and $\Pi_\gamma \psi(0) = \psi(0)$, then $\Pi_\gamma \psi(t) = \psi(t)$ for all t . Any target vector orthogonal to $\text{Ran } \Pi_\gamma$ has exactly zero transition amplitude.*

Thus changing the initial state to one adapted to another irrep, or adding a generator with nonzero inter-sector matrix elements, changes the accessible spectrum and can evade the obstruction. This is why our no-go claims are always indexed by an initial state, a generator algebra, and the symmetries preserved by the full path.

5 Dark crossings in exact symmetry quotients

If Π_a and Π_b project onto inequivalent irreps of a symmetry preserved by the entire path, then

$$\Pi_a \partial_s H(s) \Pi_b = 0. \quad (29)$$

An equality of the two branch energies is therefore a global crossing but not an avoided crossing seen by a state confined to either block.

This phenomenon is stable within the symmetry-preserving class; it is not restricted to exactly solvable diagonal cancellation points.

Theorem 3 (Stability of a transverse dark crossing). *Let $H_\varepsilon(s) = H(s) + \varepsilon V(s)$ be a continuously differentiable finite-dimensional path, and let Π_a, Π_b project onto inequivalent symmetry sectors. Suppose every $H_\varepsilon(s)$ commutes with both projectors. Assume that near $(s_\star, 0)$ the two restricted blocks have simple isolated eigenvalue branches $E_a(s, \varepsilon)$ and $E_b(s, \varepsilon)$ such that*

$$E_a(s_\star, 0) = E_b(s_\star, 0), \quad \partial_s(E_a - E_b)(s_\star, 0) \neq 0. \quad (30)$$

Then, for all sufficiently small $|\varepsilon|$, there is a unique nearby $s_\star(\varepsilon) = s_\star + O(\varepsilon)$ at which the two branches cross exactly, and

$$\Pi_a \partial_s H_\varepsilon(s) \Pi_b = 0. \quad (31)$$

If the a branch is separated from the rest of its block by at least $\Delta_\star > 0$ throughout a neighbourhood of $s_\star$ at $\varepsilon = 0$, then its isolation gap remains at least $\Delta_\star/2$ whenever $|\varepsilon| \sup_s \|V(s)\| \leq \Delta_\star/4$ and the perturbed crossing stays in that neighbourhood.

Proof. Simple isolated eigenvalues of a differentiable Hermitian family define differentiable local branches. Apply the implicit-function theorem to $F(s, \varepsilon) = E_a(s, \varepsilon) - E_b(s, \varepsilon)$ and use Eq. (30). Commutation with the symmetry projectors makes every inter-sector matrix element vanish, so the equality cannot become an avoided crossing. Finally, Weyl's bound moves each eigenvalue of the a block by at most $|\varepsilon| \|V(s)\|$; an internal spectral distance can therefore shrink by at most twice this quantity, proving the stated gap bound. $\square$

5.1 A finite joint-symmetry audit

For the frozen 2×4 incidence instance with $k = 3$, the faithful base group is $G_B \simeq D_4$ and the $S_3 \times D_4$ fixed quotient has dimension 22, compared with 512 in the valid ordered space. Cyclic closure from $|u\rangle$ also has dimension 22; this equality is an output of the closure calculation, not an assumption based on symmetry. The orbit-isometry, endpoint-invariance, and cyclic-invariance residuals are below 10^{-13} .

The uncatalysed linear path has a cancellation point at $s = 7/15$ where the fixed-space ground rank becomes two. Thus this example is also a warning: joint symmetry alone does not guarantee a nonzero accessible rank-one gap. For this finite diagnostic, let

$$W = \frac{\mathbf{1}\mathbf{1}^\top - I}{n_B - 1}, \quad C_2 = - \sum_{1 \leq r < t \leq k} W^{(r)} W^{(t)}, \quad (32)$$

and use the endpoint-preserving path

$$H_\gamma(s) = (1 - s)H_{\text{init}} + sH_{\text{prob}} + 4\gamma s(1 - s)C_2. \quad (33)$$

The operator C_2 commutes with both S_k and G_B , and its envelope vanishes at $s = 0, 1$. We set $\gamma = 2$. This catalyst moves the remaining global crossing to $s \simeq 0.806454$. At that point the two branches belong to different D_4 irreps, Eq. (29) has residual 4.7×10^{-14} , and the D_4 -trivial gap is approximately 0.258. The trivial-sector gap was evaluated on a uniform 2001-point grid and then locally minimized within every grid bracket; its minimum is 0.0414 at $s \simeq 0.874587$. The crossing between the trivial floor and the orthogonal-complement floor was bracketed on 401 points and refined by a scalar root solve. These figures are finite numerical diagnostics with residual-based degeneracy tests, not interval-certified bounds, and they are not used as an asymptotic theorem.

5.2 An exact crossing on the even-cycle family

The asymptotic witness is analytic. Express minimum vertex cover on the even cycle C_n , $n = 2k$, as set cover: bases are vertices, tasks are edges, and each base covers its two incident edges. The only minimum covers are the two alternating subsets, which form one orbit of D_n .

Consider the original ordered-register linear path

$$H(s) = (1 - s)H_{\text{init}} + sH_{\text{prob}}. \quad (34)$$

The off-diagonal coefficient of each single-register complete-graph move is

$$-\frac{1-s}{n} + \frac{s}{n-1}, \quad (35)$$

and vanishes exactly at

$$s_c = \frac{n-1}{2n-1}. \quad (36)$$

At this point the Hamiltonian is diagonal up to a scalar. All $2k!$ ordered representatives of the two alternating covers are global ground states, whereas their normalized orbit is the unique ground vector in the $S_k \times D_n$ fixed space. Assuming $\mu \geq \lambda$, the next fixed orbit violates at least one coverage constraint, and

$$g_0^{\text{global}}(s_c) = 2k!, \quad g_0^{\text{sym}}(s_c) = 1, \quad (37)$$

$$\Delta_{\text{sym}}(s_c) = \lambda \frac{n-1}{2n-1}. \quad (38)$$

Thus the global multiplicity gap closes exactly while the joint-fixed gap is bounded below by a constant. The crossing is dynamically dark for the uniform initial state and the symmetry-preserving path.

Equation (38) controls the crossing point, not the minimum gap on every part of the linear path. The next section introduces a different, explicit parent path on the same family and proves a uniform polynomial bound. Keeping these statements separate avoids turning a pointwise crossing theorem into an unsupported adiabatic claim.

6 An analytically controlled family

The distinction between a global gap and a dynamically accessible gap is not only a finite-instance effect. We now give an even-cycle family for which the symmetry reduction, the solution manifold, and an entire parent-Hamiltonian path can be controlled. The purpose of the family is a separation theorem, not a claim that vertex cover on a cycle is classically difficult.

6.1 Even-cycle set cover and its joint symmetry

Let the bases be the vertices of the even cycle C_n , $n = 2k$, and let the tasks be its edges. Base b covers the two edges incident on b . A k -subset $S \subset \mathbb{Z}_n$ is therefore a cover precisely when it is one of the two alternating subsets. The two solutions are exchanged by a one-site rotation and form one orbit of the incidence automorphism group D_n .

We work in the injective, register-symmetric subspace, with basis $\{|S\rangle : |S| = k\}$. Equivalently, this is the symmetric Dicke embedding of the hard-core sector of the original registers. Define

$$U(S) = \#\{i \in \mathbb{Z}_n : i \notin S, i+1 \notin S\}. \quad (39)$$

At half filling, $U(S)$ is also the number of adjacent occupied pairs. Thus $U = 0$ exactly on the two minimum covers.

For $a \geq 0$, consider the continuous-time Metropolis generator with a Johnson-graph proposal,

$$q_a(S, T) = \frac{1}{n} \min\{1, n^{-a[U(T)-U(S)]}\}, \quad |S \triangle T| = 2, \quad (40)$$

and $q_a(S, T) = 0$ otherwise. Its reversible measure is

$$\pi_a(S) = Z_a^{-1} n^{-aU(S)}. \quad (41)$$

The symmetric-discriminant construction is standard for reversible Markov chains and Gibbs-amplitude ground states [9, 10]. Here H_a has diagonal $\sum_T q_a(S, T)$ and off-diagonal entries

$$\langle T | H_a | S \rangle = -\sqrt{q_a(S, T)q_a(T, S)} \quad (42)$$

$$= -\frac{1}{n} n^{-a|U(T)-U(S)|/2}. \quad (43)$$

It is frustration free and has the known ground state

$$|\psi_a\rangle = Z_a^{-1/2} \sum_{|S|=k} n^{-aU(S)/2} |S\rangle. \quad (44)$$

At $a = 0$, this is the Dicke state $|J_{n,k}\rangle = \binom{n}{k}^{-1/2} \sum_{|S|=k} |S\rangle$, and the parent Hamiltonian $H_{a=0}$ is exactly the Johnson-graph Laplacian at rate $1/n$. It should not be confused with the restriction of H_{init} from Eq. (3), which agrees with a Johnson Laplacian only up to an additive constant after projection to the injective subspace. The whole parent path commutes with D_n .

This protocol has its own generator algebra and cyclic space, distinct from Eq. (12):

$$\mathfrak{A}_{\text{par}} = \text{alg}^*(\{H_a : 0 \leq a \leq 7\} \cup \{I\}), \quad (45)$$

$$\mathcal{K}_J^{\text{par}} = \mathfrak{A}_{\text{par}} |J_{n,k}\rangle. \quad (46)$$

Let $\mathcal{H}^{D_n}$ denote the D_n -fixed subspace of the half-filled hard-core space and define

$$\Delta_{\text{par}}^{\text{cyc}}(a) = \text{gap}(H_a | \mathcal{K}_J^{\text{par}}). \quad (47)$$

Proposition 4 (Cyclic accessibility of the parent path). *For every even $n = 2k \geq 6$ and $0 \leq a \leq 7$,*

1. $\mathcal{K}_J^{\text{par}}$ is a common reducing subspace for the family $\{H_a\}_{0 \leq a \leq 7}$ and $\mathcal{K}_J^{\text{par}} \subseteq \mathcal{H}^{D_n}$;
2. the unique ground state $|\psi_a\rangle$ belongs to $\mathcal{K}_J^{\text{par}}$; and
- 3.

$$\Delta_{\text{par}}^{\text{cyc}}(a) \geq \Delta_{D_n}(a). \quad (48)$$

Thus a lower bound proved in the D_n -fixed space is also a lower-bound certificate for the dynamically accessible gap of this parent protocol.

Proof. Because $\mathfrak{A}_{\text{par}}$ is a unital star-algebra, $\mathfrak{A}_{\text{par}} |J_{n,k}\rangle$ is invariant under every element of the algebra and its adjoint, hence is reducing. Every H_a commutes with D_n , and $|J_{n,k}\rangle$ is D_n -fixed, which proves the first item.

All Metropolis rates in Eq. (40) are positive on the connected Johnson graph. The zero eigenvalue of H_a is consequently simple, with ground vector given by Eq. (44). In finite dimension its spectral projector

$P_a = |\psi_a\rangle\langle\psi_a|$ is a polynomial in H_a and therefore belongs to $\mathfrak{A}_{\text{par}}$. Positivity of both vectors gives

$$\langle\psi_a | J_{n,k}\rangle > 0, \quad (49)$$

so $P_a |J_{n,k}\rangle$ is a nonzero multiple of $|\psi_a\rangle$ and the second item follows. Finally, both restrictions have this same unique ground state, while $\mathcal{K}_J^{\text{par}} \subseteq \mathcal{H}^{D_n}$. Rayleigh–Ritz minimization over the smaller orthogonal complement proves Eq. (48). $\square$

The number of configurations at energy u is

$$N_u = \frac{n}{k-u} \binom{k-1}{u}^2, \quad 0 \leq u < k. \quad (50)$$

This follows by decomposing the occupied and vacant sites into the same number $k-u$ of positive cyclic runs. At $a_f = 7$, the $u = 1$ contribution is $N_1 n^{-7} = n(k-1)n^{-7} = O(n^{-5})$. Moreover, consecutive terms $N_u n^{-7u}$ have ratio at most $2k^2 n^{-7} = O(n^{-5})$. Hence Eqs. (41) and (50) imply

$$1 - \langle\psi_{a_f} | \Pi_{U=0} | \psi_{a_f}\rangle = O(n^{-5}). \quad (51)$$

6.2 A uniform accessible-gap theorem

We first separate the cycle geometry from the zero-range kinetics.

Proposition 5 (Zero-range reduction). *For every even $n = 2k \geq 6$ and every $a \geq 0$, the gap of H_a in the D_n -fixed space obeys*

$$\Delta_{D_n}(a) \geq \frac{2}{n} [1 - \cos(2\pi/k)] \text{gap ZRP}(K_k, r_a, k), \quad (52)$$

where $r_a(0) = 0$, $r_a(1) = n^{-a}$, $r_a(j) = 1$ for $j \geq 2$, and K_k is the uniform mean-field kernel on the k boxes. Consequently, any uniform polynomial lower bound for this mean-field gap gives a uniform polynomial accessible-gap certificate for the parent path.

Proof sketch. Marking an occupied site identifies D_n -invariant subset observables isometrically with cyclic-and-reflection-invariant observables on weak compositions $g_1 + \dots + g_k = k$. Adjacent subset moves dominate a zero-range chain with cycle geometry, physical prefactor $2/n$, and rates $r_a(1) = n^{-a}$ and $r_a(j) = 1$ for $j \geq 2$. The Hermon–Salez Dirichlet-form comparison then contributes the one-particle factor $1 - \cos(2\pi/k)$. Appendix A proves the isometry, rate domination, and every normalization factor. $\square$

Lemma 6 (Three-box heat-bath contraction). *Let $a_0 = \tau$, $a_j = 1$ for $j \geq 1$, with $0 < \tau \leq 1$. On weak compositions of any total m into three boxes, let one heat-bath step keep one uniformly chosen coordinate and resample the other two from their conditional product law. The continuous-time generator $\mathcal{L}_{3,m}^*$ obtained by subtracting the identity has gap at least $1/3$.*

Proof. The log-concave coupling and its normalization are proved in Appendix A.2. $\square$

Proposition 7 (Uniform accessible gap). *For every even $n = 2k \geq 6$ and every $a \geq 0$,*

$$\text{gap ZRP}(K_k, r_a, k) \geq 2n^{-a}k^{-3}, \quad (53)$$

$$\Delta_{D_n}(a) \geq 1024 n^{-(a+6)}. \quad (54)$$

Consequently, along $0 \leq a \leq 7$, $\min_a \Delta_{D_n}(a) \geq 1024 n^{-13}$.

Proof sketch. With the normalization stated in Appendix A.3, the Caputo–Sasada recursion and Lemma 6 give heat-bath gap at least $1/k$. A conditional two-box comparison gives gap at least τ/k^2 , while the decomposition of the mean-field generator contributes the factor $2/k$. Consequently

$$\text{gap ZRP}(K_k, r_a, k) \geq 2(\tau/k^2)(1/k) = 2\tau k^{-3}. \quad (55)$$

Combining this with Proposition 5, $1 - \cos(2\pi/k) \geq 8/k^2$, and $k = n/2$ gives $1024 n^{-(a+6)}$. Appendix A.4 supplies the complete Dirichlet-form calculation. $\square$

The gap theorem can be converted into an adiabatic statement in the abstract Hamiltonian-evolution model. This does not by itself supply a gate-level implementation of the path.

Corollary 8 (Conditional adiabatic preparation). *Let $a(s) = 7s$, $0 \leq s \leq 1$, and evolve from $|J_{n,k}\rangle$ under $H_{a(s)}$ for total time T . There is a universal constant $C > 0$ such that, for every $0 < \epsilon < 1$, the sufficient condition*

$$T \geq C\epsilon^{-1}n^{41}(\log n)^2 \quad (56)$$

produces a final state at distance $O(\epsilon)$ from $|\psi_7\rangle$, up to phase. Its cover probability is therefore $1 - O(n^{-5}) - O(\epsilon)$. This runtime excludes the costs of preparing the Dicke state and realizing or simulating $H_{a(s)}$.

Proof. A Johnson move removes one selected cycle vertex and inserts one unselected vertex, so $|U(T) - U(S)| \leq 2$. Each subset has $k(n - k) = k^2$ Johnson neighbours. Differentiating the matrix elements in Eqs. (40) and (43), and bounding the operator norm by the maximum absolute row sum, gives constants c_1, c_2 independent of n such that

$$\|\partial_s H_{7s}\| \leq c_1 n \log n, \quad \|\partial_s^2 H_{7s}\| \leq c_2 n (\log n)^2. \quad (57)$$

Proposition 4 and Eq. (54) give a simple ground state and minimum cyclic gap at least $1024n^{-13}$. Substitution into a quantitative gapped adiabatic bound [11] yields contributions of orders $n^{27} \log n$, $n^{27}(\log n)^2$, and $n^{41}(\log n)^2$, divided by T . The last dominates and proves Eq. (56). The final measurement state follows from Eq. (51) and continuity of measurement probabilities in state distance. $\square$

Exact D_n -orbit quotients for $n = 6, 8, \dots, 18$ place the minimum of a 31-point grid at $a = 7$. The endpoint gaps decrease from 0.166670 at $n = 6$ to 0.0105026 at $n = 18$; the intermediate values are in the archived data table. They behave much like n^{-2} on these sizes. This is evidence about the tested grid, not a scaling theorem; Proposition 7 supplies the uniform certificate but is deliberately non-sharp.

The path also clarifies what is and is not inherited from the original multi-register Hamiltonian. Its endpoint ground state has the same solution-producing measurement objective, but the initial ground state is the injective Dicke state rather than the product-register uniform state, and the endpoint is the frustration-free Gibbs parent H_{a_f} rather than the original H_{prob} . No interpolation between these two endpoints is assumed. The exact dark crossing of the original linear interpolation and the uniformly polynomially gapped parent path are therefore complementary statements about the same symmetric instance family, not two descriptions of one protocol.

7 Numerical protocol, scope, and outlook

The numerical reconstruction follows the mathematical distinctions above. Inputs are frozen with instance identifiers and SHA-256 hashes. Every headline gap is computed either in the full Hamiltonian or in an explicitly labelled exact orbit quotient; projected *PHP*, Löwdin, and full-Hamiltonian numbers are never placed in one unlabeled column. A numerical degeneracy is accepted only relative to eigenpair residuals and matrix scale. Inside a degenerate manifold, irrep content is obtained from

$$\text{rank}(P_0 \Pi_{\alpha\beta} P_0), \quad (58)$$

not by assigning labels to solver-dependent individual eigenvectors.

7.1 Frozen endpoint reconstruction

Tables 1 and 2 implement this protocol on eleven frozen, unweighted instances. Only the 2×4 grid has nontrivial G_B (namely D_4); all other rows nevertheless retain the S_k reduction. Five rows satisfy the strict inclusion $\mathcal{K}_u \subsetneq \mathcal{H}_{\text{sym}}$, so cyclic inaccessibility is not merely a padding artefact.

At the problem endpoint, every accessible ground state is rank one in the tested sample and

$$p_{\text{opt}}^{\min} \geq 0.9944. \quad (59)$$

Nine rows have $\delta_{\text{acc}} > 0$: their uniform symmetric protocol ends above the global ground energy. In the other two rows, $\delta_{\text{acc}} = 0$ but the accessible excitation gap is orders of magnitude larger than the full gap.

Thus neither equality nor inequality of the endpoint ground energies determines which spectral gap is relevant.

For the sparse full-space calculations, the central class sum of register transpositions separates every S_k irrep for $k \leq 5$. The orbit of a lowest vector constructs its complete Specht multiplet; deflation then tests for further cospectral copies before g_0^{full} is assigned. For the grid row the two-dimensional global ground manifold is additionally the E irrep of D_4 , whereas the accessible branch is A_1 . At the analytic hopping-cancellation point, the joint-fixed ground rank agrees with the independently enumerated number of G_B -orbits of optimal covers in every row. Hence the original linear path has a fixed-sector rank closure whenever this orbit count exceeds one.

The largest orbit-invariance residual is below 5×10^{-13} , the largest cyclic-invariance residual below 2×10^{-9} , and all full-space eigenpair and class-sum residuals used for reported levels are stored in the machine-readable output. The table values never combine projected *PHP*, Löwdin, and full-Hamiltonian estimates.

The theorem family establishes a qualitative separation, not a practical speedup. Its cycle instances are classically solvable, Dicke-state preparation has a cost that must be counted, and the Gibbs parent endpoint is not the original problem Hamiltonian. Likewise, the address count $k \lceil \log_2 n_B \rceil$ is not a complete hardware resource estimate: incidence access, flags, ancillas, and gate depth belong to a separate implementation analysis.

The next nontrivial question is stability: one would like a decorated family in which the classical cover structure is not immediate, while the joint orbit quotient, dark global branch, and polynomial accessible gap survive controlled symmetry-preserving perturbations. The present contribution should be read as a rigorous theory of spectral relevance under joint symmetry, with an exactly controlled dark crossing, a polynomially gapped witness path, and reproducible finite-instance diagnostics.

A Normalized comparison proof for the parent-path gap

This appendix fixes the generator convention used in Propositions 5 and 7 and records each comparison factor. For a reversible generator $\mathcal{L}$ with stationary law ν , write

$$\mathcal{E}_{\mathcal{L}}(f, f) = -\langle f, \mathcal{L}f \rangle_{\nu}, \quad (60)$$

$$\text{gap}(-\mathcal{L}) = \inf_{\text{Var}_{\nu}(f) > 0} \frac{\mathcal{E}_{\mathcal{L}}(f, f)}{\text{Var}_{\nu}(f)}. \quad (61)$$

A.1 Marked configurations and the cycle zero-range chain

Put $\tau = n^{-a}$ and let $\Omega_{n,k} = \{S \subset \mathbb{Z}_n : |S| = k\}$. The Markov generator whose symmetric discriminant is H_a has rates from Eq. (40) and stationary law $\pi_a(S) \propto \tau^{U(S)}$. If D_{π_a} is diagonal with entries $\pi_a(S)$, then

$$H_a = -D_{\pi_a}^{1/2} \mathcal{Q}_a D_{\pi_a}^{-1/2}. \quad (62)$$

The unitary map $f \mapsto \sum_S \sqrt{\pi_a(S)} f(S) |S\rangle$ intertwines the D_n actions. Thus the gap of H_a in the fixed space is the Poincaré constant of $\mathcal{Q}_a$ on D_n -invariant observables.

Introduce the marked space

$$\hat{\Omega}_{n,k} = \{(S, x) : S \in \Omega_{n,k}, x \in S\}, \quad (63)$$

$$\hat{\pi}_a(S, x) = \frac{\pi_a(S)}{k}. \quad (64)$$

Starting at the marked particle x , list the occupied vertices in cyclic order and let g_i be the number of vacant vertices after particle i . This gives a bijection between marked configurations and anchored weak compositions satisfying

$$g_i \geq 0, \quad \sum_{i=1}^k g_i = k, \quad U(g) = \#\{i : g_i = 0\}. \quad (65)$$

Changing the marked particle cyclically shifts g , while rotating or reflecting the subset changes only the anchor and the dihedral orientation. Consequently a D_n -invariant observable $f(S)$ lifts to a cyclic-and-reflection-invariant observable $F(g)$.

Define

$$r_a(0) = 0, q\text{quadr}_a(1) = \tau, q\text{quadr}_a(j) = 1 \quad (j \geq 2). \quad (66)$$

The product weight of a composition is

$$\prod_{i=1}^k \prod_{j=1}^{g_i} r_a(j)^{-1} = \tau^{-\#\{i: g_i > 0\}} = \tau^{-k} \tau^{U(g)}. \quad (67)$$

The irrelevant factor τ^{-k} shows that the pushforward of the marked law is exactly the stationary composition law. In particular,

$$\text{Var}_{\pi_a}(f) = \text{Var}_{\hat{\pi}_a}^{\wedge}(f) = \text{Var}_{\nu_a}(F). \quad (68)$$

Let $P_{C_k}(i, j) = 1/2$ when $j = i \pm 1 \pmod{k}$ and zero otherwise. We use the Hermon–Salez convention

$$(\mathcal{L}_{P,r} F)(g) = \sum_{i,j} r(g_i) P(i, j) [F(g - e_i + e_j) - F(g)]. \quad (69)$$

Multiplication by $2/n$ makes every oriented cycle transfer occur at rate $r_a(g_i)/n$. Such a transfer is an adjacent Johnson move. If $g_i = 1$ and the receiving box is positive, it creates one new zero and has the Metropolis rate τ/n ; if the receiving box is zero, its

Table 1: Exact structural reduction for the frozen finite instances. All dimensions refer to the physical, unpadding address space.

instance	k	$ G_B $	$\dim \mathcal{H}_{\text{valid}}$	$\dim \mathcal{H}_{\text{sym}}$	$\dim \mathcal{K}_u$	cover orbits
grid-2x4	3	8	512	22	22	2
rnd-b8-t7-p50-s002	3	1	512	120	101	12
rnd-b8-t8-p42-s014	4	1	4096	330	326	7
scpe1-b8-t20	4	1	4096	330	328	8
scpe1-b8-t30	5	1	32768	792	792	1
scpe2-b8-t20	4	1	4096	330	328	3
scpe2-b8-t30	5	1	32768	792	792	8
scpe3-b8-t20	3	1	512	120	119	2
scpe3-b8-t30	5	1	32768	792	792	4
scpe4-b8-t29	5	1	32768	792	792	1
scpe5-b8-t30	5	1	32768	792	792	2

Table 2: Problem-endpoint spectra reconstructed from the full Hamiltonian and the exact cyclic restriction. Here Δ is the gap above the entire ground manifold, and $p_{\text{opt}}^{\min}$ is minimized over normalized accessible ground states. The last column is the joint-fixed ground rank at the analytic hopping-cancellation point.

instance	ground S_k content	g_0^{full}	Δ_{full}	δ_{acc}	g_0^{acc}	Δ_{acc}	$p_{\text{opt}}^{\min}$	$g_0^{\text{sym}}(s_c)$
grid-2x4	2[3]	2	0.000382	0.271	1	0.315	0.9944	2
rnd-b8-t7-p50-s002	$[3] \oplus 2[2, 1] \oplus [1, 1, 1]$	6	0.0563	0.0581	1	0.128	0.9986	12
rnd-b8-t8-p42-s014	$[2, 1, 1]$	3	0.000327	0.0274	1	0.0297	0.9955	7
scpe1-b8-t20	$[1, 1, 1, 1]$	1	0.000539	0.0103	1	0.00178	0.9983	8
scpe1-b8-t30	[5]	1	0.00591	0	1	4.87	0.9949	1
scpe2-b8-t20	$[2, 1, 1]$	3	0.00137	0.00674	1	0.18	0.9966	3
scpe2-b8-t30	$[1, 1, 1, 1, 1]$	1	0.00118	0.0316	1	0.0829	0.9982	8
scpe3-b8-t20	$[2, 1]$	2	0.000658	0.00499	1	0.256	0.9969	2
scpe3-b8-t30	$[3, 1, 1]$	6	0.000434	0.00864	1	0.247	0.9985	4
scpe4-b8-t29	[5]	1	0.00608	0	1	4.8	0.9945	1
scpe5-b8-t30	$[4, 1]$	4	0.00306	0.00749	1	0.242	0.9962	2

Metropolis rate is $1/n \geq \tau/n$. For $g_i \geq 2$, the move does not increase U and both rates equal $1/n$. Hence the full Johnson Dirichlet form dominates the local form and, under the lift above,

$$\mathcal{E}_{\mathcal{Q}_a}(f, f) \geq \frac{2}{n} \mathcal{E}_{\text{ZRP}(P_{C_k, r_a, k})}(F, F). \quad (70)$$

The equality of variances is exact; the inequality comes only from dropping nonlocal Johnson edges and, in one local case, lowering a Metropolis rate.

Let $K_k(i, j) = 1/k$. The Hermon–Salez comparison theorem [12] states at the level of Dirichlet forms that

$$\begin{aligned} \mathcal{E}_{\text{ZRP}(P_{C_k, r_a, k})}(F, F) \\ \geq [1 - \cos(2\pi/k)] \mathcal{E}_{\text{ZRP}(K_k, r_a, k)}(F, F), \end{aligned} \quad (71)$$

because $1 - \cos(2\pi/k)$ is the Poincaré constant of P_{C_k} and that of K_k is one. Restricting the variational problem to dihedral-invariant F can only increase its infimum. Equations (68)–(71) prove Eq. (52) with its factor $2/n$.

A.2 Three-box contraction

Set $a_0 = \tau$ and $a_j = 1$ for $j \geq 1$. For $0 < z < 1$, the probability sequence $p_z(j) \propto a_j z^j$ is log-concave: the only nontrivial inequality is $a_1^2 \geq a_0 a_2$, which is precisely $\tau \leq 1$. Conditioning two independent variables with this law to have sum s removes the factor z^s and gives

$$\nu_s(u) = \frac{a_u a_{s-u}}{\sum_{v=0}^s a_v a_{s-v}}, \quad 0 \leq u \leq s. \quad (72)$$

The discrete Efron monotonicity theorem [13] implies that $U_s \sim \nu_s$ is stochastically nondecreasing in s . Symmetry under $u \mapsto s - u$ gives, for $t = s + d$,

$$U_s \preceq_{\text{st}} U_t \preceq_{\text{st}} U_s + d. \quad (73)$$

The common-quantile coupling therefore obeys $0 \leq U_t - U_s \leq d$ almost surely. The two increments in the coupled splits are $U_t - U_s$ and $d - (U_t - U_s)$, so both are nonnegative.

On the three-box state space use $d_1(x, y) = \frac{1}{2} \sum_i |x_i - y_i|$. Couple two heat-bath steps by holding the same uniformly selected coordinate i . The two

resampled pair totals differ by $d = |x_i - y_i|$, and the coupling above gives pairwise ℓ^1 distance d . Together with the held-coordinate distance, the post-update d_1 distance is at most $|x_i - y_i|$. Thus the discrete heat-bath kernel $\mathbf{P}$ satisfies

$$W_{1,d_1}(\mathbf{P}(x, \cdot), \mathbf{P}(y, \cdot)) \leq \frac{1}{3} \sum_i |x_i - y_i| = \frac{2}{3} d_1(x, y). \quad (74)$$

It follows that $\text{Lip}(\mathbf{P}f) \leq (2/3) \text{Lip}(f)$. Every non-constant eigenfunction on this finite metric space has positive Lipschitz seminorm, so all nonconstant eigenvalues of the reversible kernel have absolute value at most $2/3$. Therefore $\mathcal{L}_{3,m}^* = \mathbf{P} - I$ has gap at least $1/3$, uniformly in the conserved total m . This proves Lemma 6.

A.3 Caputo–Sasada recursion with the present normalization

For N boxes and conserved total m , let

$$\Omega_{N,m} = \{g \in \mathbb{Z}_{\geq 0}^N : \sum_i g_i = m\}, \quad (75)$$

$$\nu_{N,m}(g) = Z_{N,m}^{-1} \prod_{i=1}^N a_{g_i}. \quad (76)$$

Let E_{ij} denote conditional expectation under $\nu_{N,m}$ given all coordinates outside the unordered pair $\{i, j\}$. The complete-graph pair heat-bath generator is normalized as

$$\mathcal{L}_{\text{HB}}^{N,m} = \frac{1}{N} \sum_{1 \leq i < j \leq N} (E_{ij} - I). \quad (77)$$

For $N = 3$, the discrete kernel $I + \mathcal{L}_{\text{HB}}^{3,m}$ keeps one uniformly chosen coordinate and resamples the remaining pair. It is exactly the kernel of Lemma 6, not a rescaled version.

The conditioned product measures above have the non-interference property required by the three-component argument. With

$$\lambda_3 = \inf_{m \geq 0} \text{gap}(-\mathcal{L}_{\text{HB}}^{3,m}), \quad (78)$$

the lower-bound form of the Caputo–Sasada recursion [14, 15], in the normalization (77), is

$$\text{gap}(-\mathcal{L}_{\text{HB}}^{N,m}) \geq (3\lambda_3 - 1) \left(1 - \frac{2}{N}\right) + \frac{1}{N}. \quad (79)$$

Lemma 6 gives $\lambda_3 \geq 1/3$. The first term on the right of Eq. (79) is therefore nonnegative, and

$$\text{gap}(-\mathcal{L}_{\text{HB}}^{N,m}) \geq \frac{1}{N} \quad (80)$$

for every conserved total. In the application below, $N = m = k$.

A.4 Conditional two-box comparison

For the mean-field kernel $K_k(i, j) = 1/k$, decompose its zero-range generator into unordered pairs:

$$\mathcal{L}_{\text{ZRP}(K_k, r_a, m)} = \frac{2}{k} \sum_{1 \leq i < j \leq k} \mathcal{L}_{ij}^{(2)}. \quad (81)$$

Here $\mathcal{L}_{ij}^{(2)}$ is the conditional two-box zero-range generator with kernel K_2 , whose two rows are $(1/2, 1/2)$. Thus a unit leaves either member of the pair toward the other at rate $r_a/2$. The prefactor $2/k$ in Eq. (81) converts this to the mean-field rate r_a/k .

Condition on pair total s . For $s \geq 2$, the stationary weights on $\{0, \dots, s\}$ are τ at the two endpoints and one in the interior, with

$$Z_s = s - 1 + 2\tau. \quad (82)$$

Every nearest-neighbour edge conductance of $\mathcal{L}_{ij}^{(2)}$ is at least $\tau/(2Z_s)$. For the canonical path between $u < v$, Cauchy–Schwarz gives a factor $v - u \leq s$. For every edge, the stationary masses on its two sides have product at most $1/4$. Consequently

$$\text{Var}_{\nu_s}(f) \leq \frac{s}{4} \sum_{u=0}^{s-1} [f(u+1) - f(u)]^2 \leq \frac{sZ_s}{2\tau} \mathcal{E}_s(f, f). \quad (83)$$

The conditional pair gap is at least

$$\frac{2\tau}{sZ_s} \geq \frac{2\tau}{s(s+1)} \geq \frac{\tau}{k^2}, \quad 1 \leq s \leq k; \quad (84)$$

for $s = 1$ it equals τ , and $s = 0$ has no conditional variance.

Let $\mathcal{E}_{\text{HB}}$ be the form of Eq. (77). Applying Eq. (84) conditionally in every pair and using Eq. (81) gives, with $\mathcal{V}_{ij}(f)$ denoting the expected conditional variance outside $\{i, j\}$,

$$\mathcal{E}_{\text{ZRP}(K_k, r_a, k)}(f, f) \geq \frac{2}{k} \frac{\tau}{k^2} \sum_{i < j} \mathcal{V}_{ij}(f) \quad (85)$$

$$= \frac{2\tau}{k^2} \mathcal{E}_{\text{HB}}(f, f). \quad (86)$$

Combining this with Eq. (80) yields

$$\text{gap ZRP}(K_k, r_a, k) \geq \frac{2\tau}{k^2} \frac{1}{k} = 2\tau k^{-3}. \quad (87)$$

This displays separately the pair-selection factor $2/k$, the conditional pair estimate τ/k^2 , and the heat-bath gap $1/k$.

Finally, for $k \geq 3$, $1 - \cos(2\pi/k) \geq 8/k^2$. Substitution of Eq. (87) into Eq. (52) gives

$$\Delta_{D_n}(a) \geq \frac{2}{n} \frac{8}{k^2} \frac{2\tau}{k^3} = \frac{32\tau}{nk^5} = 1024 n^{-(a+6)}, \quad (88)$$

because $k = n/2$ and $\tau = n^{-a}$. This completes the proof of Proposition 7.

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Code and data availability

The frozen instance registry, source code, residual-based numerical checks, and machine-readable tables will be deposited in an archived, versioned repository before submission. No placeholder URL will appear in the submitted manuscript. Original software is released under the MIT License; original data and documentation are released under CC BY 4.0. Third-party materials, including OR-Library incidence data and the journal class file, retain their source licenses and provenance.

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